Week 1 (May 5–11) — Project Planning + Research

Goal: Understand the project deeply before coding.

Tasks:

- Research Reddit/news sentiment systems
- Create Flowchart and start on Gantt + Pert charts
- Decide project architecture
- Decide what APIs/data sources you will use
- Create GitHub repository
- Organize project folders
- Create a weekly plan

Deliverables:

- detailed flowchart
- GitHub repo setup
- Project outline

Week 2 (May 12–18) — Gantt &Pert charts + System Design

Goal: Start building the project foundation from the completed flowchart.

Tasks

- Revise flowchart based on professor feedback
- Finalize Gantt chart + PERT chart
- Set up development environment
- Connect/test APIs:
  - Reddit API
  - news APIs
- Begin collecting sample stock discussion data
- Test storing data into CSV/JSON/database
- Organize GitHub folders:
  - data/
  - scripts/
  - models/
  - dashboard/
  - docs/
- Improve README setup instructions
- Begin documenting workflow steps

Week 3 (May 19–25) — Data Collection Setup

Goal: Start building the data pipeline.

Tasks:

- Connect Reddit API/news APIs
- Test pulling:
    - ticker mentions
    - post titles
    - comments
    - timestamps
- Save raw data into CSV/JSON/database
- Create scripts for automated data collection

Deliverables:

- Working API scripts
- Sample collected datasets
- GitHub commits with comments

Week 4 (May 26–June 1) — Data Cleaning & Preprocessing

Goal: Turn messy text into usable data.

Tasks:

- Remove spam/noise
- Normalize text
- Handle emojis/symbols
- Extract stock tickers
- Tokenization/preprocessing
- Build clean dataset pipeline

Deliverables:

- Clean preprocessing module
- Before vs after cleaned data examples
- organized GitHub
- flowcharts
- actual working code
- weekly progress

Week 5 (June 2–8) — Sentiment Analysis Development

Goal: Build the sentiment engine.

Tasks:

- Research sentiment models/libraries
- Implement:
    - positive
    - negative

    - neutral scoring
- Test accuracy on sample posts
- Compare methods

Deliverables:

- Working sentiment classifier
- Example outputs

Week 6 (June 9–15) — Stock Discussion Analytics

Goal: Generate useful metrics.

Tasks:

- Count ticker mentions
- Rank trending stocks
- Measure discussion frequency
- Calculate sentiment averages
- Build summary analytics

Deliverables:

- Trend analysis outputs
- Ranked ticker lists

Week 7 (June 16–22) — Backend Integration

Goal: Connect all modules together.

Tasks:

- Combine:
    - API collection
    - preprocessing
    - sentiment analysis
    - analytics
- Create automated workflow pipeline
- Add logging/error handling

Deliverables:

- End-to-end working pipeline

Week 8 (June 23–29) — Dashboard Design

Goal: Start visualization system.

Tasks:

- Choose dashboard framework:
    - Streamlit
    - Flask
    - Dash
- Design layout
- Add charts/tables

Deliverables:

- Initial dashboard prototype

Week 9 (June 30–July 6) — Dashboard Features

Goal: Improve interactivity.

Tasks:

- Real-time updates
- Search/filter ticker
- Sentiment graphs
- Mention-density charts
- UI improvements

Deliverables:

- Functional dashboard

Week 10 (July 7–13) — Testing & Debugging

Goal: Make the project reliable.

Tasks:

- Fix bugs
- Test on different machines
- Improve README setup instructions
- Improve code comments

Deliverables:

- Stable project version

Week 11 (July 14–20) — Optimization + Documentation

Goal: Professionalize the project.

Tasks:

- Refactor messy code
- Improve folder structure
- Add modular functions/classes
- Improve GitHub organization
- Add screenshots/examples

Deliverables:

- Professional GitHub repo

Week 12 (July 21–27) — User Manual Recording Prep

Goal: Prepare presentation/demo.

Tasks:

- Script demo flow
- Prepare example use cases
- Practice explaining dashboard
- Prepare webcam presentation

Deliverables:

- Draft Kaltura demo outline

Week 13 (July 28–August 2) — FINAL DEVELOPMENT DEADLINE

Goal: Finish EVERYTHING technical.

Tasks:

- Complete all coding
- Final testing
- Final charts/dashboard
- Final GitHub cleanup

Deliverables:

- Fully completed project

Week 14 (August 3–9) — Recordings

Demo Recording (min 15, max 1h)

Present internship work since the beginning of May 5

Technical Recording:

Points to talk about:

1) The theoretical mindset you had to complete your project.

2) Explain your codes one codeset at a time.

3) Run the code and show its intended output.

This recorded session acts as a tutorial on implementation of your project.

Week 15 (August 10–15) — Final Presentation + Wrap-Up

Goal:
Finish professionally.

Tasks:

- Final user-manual presentation
- Final documentation review
- Submit all links/files
- Final GitHub verification

Deliverables:

- Final Kaltura presentation
- Final GitHub repo
- Final internship submission